AAE 512 Computational Aerodynamics

Spring 2026
Credit: 3

Course Description:

This course introduces finite-difference (FD) and finite volume (FV) methods in computational fluid dynamics (CFD). The course is divided into three parts. Part 1 presents the building blocks needed to develop, analyze, and implement CFD, including methods for initial and boundary-value problems, methods for linear and nonlinear algebraic equations, classification and properties of partial differential equations (PDEs), and the equations that govern fluid mechanics, heat transfer, and combustion problem and their underlying assumptions. Part 2 presents FD and FV methods, showing how the building blocks are assembled and their limitations. These include governing equations in generalized coordinates, grid generation, difference operators, and methods for analyzing consistency, stability, convergence, and errors such as conservative, transportive, dissipative, and dispersive as well as aliasing, monotonicity, and positivity. Part 3 shows how FD and FV methods are applied to the Euler and the Navier-Stokes equations for compressible and incompressible flows with correct implementation of boundary conditions.

Course Information

- **Instructor:** Tom I-P. Shih, Professor of Aeronautics and Astronautics
  - E-mail: tomshih@purdue.edu (best way to reach me)
  - Office Phone: (765) 494-5118 (call if urgent)
- **Teaching Assistant:**
  - Dylan Hsieh, PhD student - E-mail: hsieh90@purdue.edu; Office Hours (to handle grading issues only): TBA
  - Ayush Agarwal, MS student - E-mail: agarw508@purdue.edu; Office Hours (to handle grading issues only): TBA
- **Graders:**
  - Walter Richardson, MS student; E-mail: richa541@purdue.edu; Office Hours (to handle grading issues only): TBA
- **Office Hours:**
  - **11:30 am – 1 pm** on Mondays **at ARMS 1028**
  - **8-10 pm on Mondays and 9-10 pm on Wednesdays** for EPE and on-campus students via ZOOM at https://purdue-edu.zoom.us/j/7740731669?pwd=aS9OMDVTazI2aVZWVFlmYjl5YjhYUT09.
  - other times by appointment
- **Piazza:** Setup by TA and available. Please ask questions and have discussion there. TAs and/or instructor will respond when needed.
- **Brightspace:** syllabus, lecture notes, homework assignments, and handouts can be found on Brightspace. The materials on Brightspace are only for those enrolled in the class. The lectures, HW, and exams are copyrighted. Do not share, upload, or sell.
- **Prerequisites:**
  - Undergraduate courses in thermodynamics, fluid mechanics, and heat transfer.
  - Some background in gas dynamics, numerical methods, and a programming language such as Fortran, C, or MATLAB.
  - AAE 412 for undergraduate students from Purdue unless with instructor permission.

Class Time and Location

- **Class Time:** 10:30 - 11:20 am on MWF
- **Class Location:** Wang 2555
- **Class Attendance**:
  - Required for on-campus students (-1 point on average of final grade for every unexcused lecture missed). Why? If a student misses or does not understand concepts presented in an earlier lecture, then he/she will not understand the next lecture. Would like for students to understand the material of each lecture before attending the next lecture by asking questions in class and via Piazza and by attending office hours.
  - For online students, attendance is not required, but suggest keeping up with the lecture material (e.g., watch the recorded version during the day of the lecture).

Course Material

- **Textbook:** Jameson, A., *Computational Aerodynamics*, Cambridge University Press, 2022. Can access online at Permalink:
  https://purdue.primo.exlibrisgroup.com/discovery/fulldisplay?context=L&vid=01PURDUE_PUWL:PURDUE&docid=alma99170389856501081\
- **Lecture notes:** provided on Brightspace.
- **References:** provided at the end of this syllabus after course outline.

Course Outline:

- Given at the end of this syllabus.

Course Learning Objectives:

- Be able to develop algorithms for obtaining solutions to ordinary and partial differential equations (ODEs ad PDEs) encountered in aerospace and mechanical engineering with appropriate initial and boundary conditions.
- Understand and be able to assess sources of errors incurred by numerical methods for PDEs and how spurious solutions could be minimized or eliminated.
- Be able to properly perform a CFD study of a fluid-flow problem by using a research or a commercial CFD code from governing equations, domain size, initial and boundary conditions, and grid to verification and validation.

Course Grade:

- Exam 1: Friday, March 6, 2026 (in class) 10%
- Exam 2: Friday, May 1, 2026 (in class) 10%
- no final exam
- Homework: due by 10 am on Fridays 70%
- Project: due by midnight on last day of classes 10%

Project: Develop a 2-D CFD code based on the "thin-layer" equations and generate solutions for compressible flow in a nozzle with heat transfer.

Grade Scale:

A = 90 – 100, A- = 85-90, B+ = 80-84, B = 70 – 79, B- = 65 – 69, C = 50 – 64, D = 40-50, F < 40

Policy on Homework

- Homework will be assigned every other week on Fridays (with a few exceptions).
- All "completed" homework must be submitted to Gradescope and is due by 10 am on the due date.
- No late HW will be accepted. If there are extenuating circumstances, then please let me know ASAP via e-mail.
- Solutions of homework will be made available once all HW have been turned in, and you are encouraged to check the solutions to understand not just the solution procedures, but also interpretation of results as well as alternate approaches.
- Homework may be done in collaboration to the point of understanding the problem, problem formulation, and method of solution. Discussion is encouraged. However, each student must do his/her own programming, post processing, analysis, and writing independently. Violation will be considered cheating and reported. For assignments that involve programming, the computer programs written must be submitted along with the algorithm and flow chart used to write the program. This is an important part of the course, and each student is encouraged to spend the time to debug his/her codes until they work.

  PLEASE DO NOT wait until the day or two before the HW is due to get clarification or help from the instructor or the TA. There is no way to finish the HW one or two days before it is due. The HW is designed to help you learn the material. Please start on the HW once assigned to see what you need to understand. Office hours are designed to help with the HW due in 2 weeks. **Note that the instructor's office hour is on Monday to ensure that you get started right away. You can also ask questions about the HW during class.**

Policy on the Use of AI

- This course allows and encourages students to use AI tools for HW and class discussions but not for exams. When students use AI tools on HW, must reference the AI tool used to not violate plagiarism.
- Since the information provided by AI tools is not vetted, it could be wrong or incomplete. Thus, it is the student's responsibility to vet the "AI-provided info" used in the submitted HW.

How to succeed in this course?

- Read lecture notes before coming to class.
- Ask questions during lectures to make each lecture useful.
- If you still do not understand, then study more by re-reading lectures, references, and attend office hours to ask questions.
- Keep up so that you can understand the lectures. Otherwise, the lectures will be a waste of time.
- Read the HW assignment once available and think about how you would approach it. This is the first step to start on the HW in gauging what you still need to understand before you can do it and how long it will take you to do it.
- The most important part of education is to provide the knowledge, the skills, and the ability to self- learn so that life-long learning is possible. This course is intended to give you the KSA for self-learning in numerical methods and CFD.

Access to Fortran

- AAE students can get access to our department computers/servers to execute "GNC Fortran (gfortran)" or "intel Fortran (ifort)" by loading the following modules:
  - *$ module load gcc/7.1.0*
  - *$ module load intel/18.0.1*
- If students want to run Fortran on their PCs (Windows/Linux/Mac), suggest installing an IDE (Code::Blocks) via the website below: https://www.codeblocks.org/

Academic Integrity

- Students are reminded of the **Student Code of Honor** that states: The purpose of the Purdue University academic community is to search for truth and to endeavor to communicate with each other. Self-discipline and a sense of social obligation within each individual are necessary for the fulfillment of these goals. It is the responsibility of all Purdue students to live by this code, not out of fear of the consequences of its violation, but out of personal self-respect. As human beings we are obliged to conduct ourselves with high integrity. As members of the civil community we have to conduct ourselves as responsible citizens in accordance with the rules and regulations governing all residents of the state of Indiana and of the local community. As members of the Purdue University community, we have the responsibility to observe all University regulations.
- To foster a climate of trust and high standards of academic achievement, Purdue University is committed to cultivating academic integrity and expects students to exhibit the highest standards of honor in their scholastic endeavors. Academic integrity is essential to the success of Purdue University's mission. As members of the academic community, our foremost interest is toward achieving noble educational goals and our foremost responsibility is to ensure that academic honesty prevails."
- Please visit http://www.purdue.edu/univregs/pages/stu_conduct/code_of_honor.html.
- Academic dishonesty will result in a zero for that assignment/exam and may result in a failing grade for the course. Furthermore, for all cases of academic dishonesty, a report will be submitted to the Office of the Dean of Students, which may result in further disciplinary action. For more information go to: http://www.purdue.edu/odos/osrr/.
- All exams in this course are open book & notes; BUT, no internet, phone, calculator, and any communications with others. Also, all must sign the following **HONOR PLEDGE:** "I did not receive help or give help on this exam." Honor is the only thing that we can take with us when we die. Thus, treasure it. Once we lose it, it is hard to get it back.

Nondiscrimination

The University statement on the subject can be found at (https://www.purdue.edu/home/ea_eou_statement/). The first paragraph of that statement is "*Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his*

or her own potential. The University believes that intellectual and cultural diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life".

In the Event a Student is Quarantined

If a student becomes sick and can no longer do the work and keep up with the class, then the student can request a grade of "incomplete" that can be made up when the student recovers.

In the Event of a Campus-Wide Emergency

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances." Information about changes in this course (AAE 512) will be made available via announcements on Purdue BrightSpace and via e-mail.

References on CFD:

- Pletcher, R.H., Tannehill, J.C., and Anderson, D.A., *Computational Fluid Mechanics and Heat Transfer*, 3rd Edition, Taylor & Francis Book Company, Washington, DC, 2012.
- Ferziger, J.H. and Peric, M., *Computational Methods for Fluid Dynamics*, Springer, Berlin, 1999.
- Hirsch, C., *Numerical Computation of Internal and External Flows*, Vol. 1, John Wiley and Sons, Chichester, 1988.
- Hirsch, C., *Numerical Computation of Internal and External Flows*, Vol. 2, John Wiley and Sons, Chichester, 1990.
- Shyy, W., Udaykumar, H.S., Rao, M.M., and Smith, R.W., *Computational Fluid Dynamics with Moving Boundaries*, Taylor & Francis, Washington, D.C.,1996.
- Carey, G.F., *Computational Grids: Generation, Adaptation, and Solution Strategies*, Taylor & Francis, Washington, DC, 1997.
- Roache, P.J., *Verification and Validation in Computational Science and Engineering*, Hermosa Publishers, New Mexico, 1998.
- Oberkampf, W.L. and Roy, C.J., *Verification and Validation in Scientific Computing*, Cambridge University Press, Cambridge, 2010.
- Shih, T.I-P. & Caughey, D.A., *Foundations of Computational Fluid Dynamics*, in preparation (copyrighted by Tom Shih & Dave Caughey and only for use by students in AAE 512).

References on Numerical Methods:

- Chapra, S.C. and Canale, R.P., *Numerical Methods for Engineers*, 3rd edition, WCB/McGraw-Hill, Boston, 1998.
- Hornbeck, R.W., *Numerical Methods*, Prentice-Hall, Englewood Cliffs, New Jersey1975.
- Isaacson, E. and Keller, H.B., *Analysis of Numerical Methods*, John Wiley and Sons, New York, 1966.
- Dahlquist, G. and Bjorck, A., *Numerical Methods*, Prentice-Hall,Englewood Cliffs, N.J., 1974.
- Ralston, A. and Rabinowitz, P., *A First Course in Numerical Analysis*, Second Edition, McGraw-Hill Book Co., New York, 1978.

References on Turbulence Modelling:

- Cebeci, T. and Smith, A.M.O., *Analysis of Turbulent Boundary Layers*, Academic Press, 1974.
- Wilcox, D.C., *Turbulence Modeling for CFD*, DCW Industries, Inc., La Canada, CA, 1993.
- Pope, S., *Turbulent Flows*, Cambridge University Press, 2000.
- Durbin, P. and Pettersson-Reif, B. A., *Statistical Theory and Modeling for Turbulent Flow*, 2nd ed., John-Wiley & Sons: Chichester, 2010.

Reference on Combustion:

- Williams, F.A., *Combustion Theory*, 2nd Edition, 1985.
- Kuo, K.K., *Principles of Combustion*, Wiley, New York, 1986.

COURSE OUTLINE
AAE 512 Computational Aerodynamics

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1. Preliminaries
- What is CFD?
- What does it involve?
- Course Objectives

2. Building Blocks of CFD
- Initial-Value Problems
- Boundary-Value Problems (linear & nonlinear)
- Matrices, Matrix Properties, Eigenvalues, and Norms
- Systems of Linear and Nonlinear Equations

3. Introduction to PDEs
- Classification of PDEs (single & systems of 1st and 2nd-order linear and quasilinear PDEs with up to 4 independent variables – x, y, z, & t)
- Mathematical Properties (appropriate initial and boundary conditions)
- Genuine and Weak Solutions, Entropy Conditions

4. Governing Equations
- Assumptions and Implications/Consequences of Assumptions
- Continuity, Species Balance, Linear Momentum, and Energy
- Compressible and Incompressible Euler and Navier-Stokes with and without Chemical Reactions
- Model Equations
- An Introduction to Turbulence Modelling

5. Introduction to Finite-Difference (FD) & Finite-Volume (FV) Methods
- Overview of Some Simple FD and FV Methods via Model Equations
- Numerical Connection between Parabolic, Hyperbolic, and Elliptic PDEs (time marching vs. iteration)

6. Consistency, Stability, and Convergence Analysis
- Overview of Consistency, Stability, and Convergence
- Conditional and Unconditional Consistency
- Discrete Perturbation, Fourier, and Matrix Methods
- The Modified-Equation Method, CFL Condition
- Extension of Linear Theory to PDEs with Nonlinearities and Unbounded Solutions
- Lax-Equivalence Theorem

7. Error Analysis
- Conservation, Transportive, Dissipation, Dispersion, Anisotropy, Monotonicity, Positivity
- Aliasing Errors in Numerical Solutions
- Spatial Resolution of Mesh and Mesh+FDEs/FVEs
- Error Estimation via the Discrete-Error Transport Equations (DETE)
- Uncertainty Quantification

Exam 1: March 6, 2026

8. Grid Generation
- Basic Concepts
- Generalized Coordinates and Geometric Conservation Laws
- Finite-Volume Interpretation of Generalized Coordinates
- FD & FV Methods for an Arbitrary Geometry Using Structured Grids
- Overview of Grid Generation Methods
- Algebraic Grid-Generation Methods (stretching function, transfinite interpolation)
- Grid Topology and Domain Decomposition for High-Quality Grids and Fast Convergence Rate
- An Intro to Grid Generation Based on PDEs (elliptic, parabolic, and hyperbolic PDEs)
- An Intro to Unstructured Grids and Solution Procedure on Unstructured Grids

9. Methods for Deriving Difference Operators for FDMs or Fluxes for FVMs

- Generalized Time-Differencing Formulas and ODE-IVP Formulas
- Taylor's Series Methods, Compact Differencing (Pade Operators)
- Methods Based on Solutions of Model Equations (exponential, hybrid, power, Riemann, and approximate Riemann such as flux-vector and flux-difference schemes, limiters)
- An Introduction to High-Order Methods

10. Methods for Developing Solution Algorithms

- Space-Time Coupled Methods vs Space-Time Decoupled Methods (the method of Lines)
- Steady-State vs Time-Accurate Methods
- Explicit Methods (Predictor-Corrector, LOD, Time-Split, ADE)
- Implicit Methods (ADI, LU, approximate factorization, point and line relaxation methods)
- Hybrid Methods (explicit, implicit)
- Methods to Accelerate Convergence Rate (e.g., multi-grid methods)

11. Methods for Compressible Viscous & Inviscid Flows (Euler & Navier-Stokes)

- Implicit, Explicit, Hybrid
- Boundary Conditions (inflow and outflow)
- Pre-conditioning for Low-Mach Number Flows

12. Methods for Incompressible Flows (Navier-Stokes)

- Pressure-Based Methods (SIMPLE, SIMPEC, …)
- Methods Based on Artificial Compressibility

Exam 2: May 1, 2026